

Off-center ball movements --- Solution

X24 (2024) — English translation

- A1** Express the velocity component $\vec{u}_A$ of point A in terms of the velocity component $\vec{u}_C$ of the center of the ball, its angular velocity $\vec{\omega}$, as well as the radius vector $\vec{r}$ at an arbitrary moment. Also obtain the time derivative $\dot{\vec{u}}_A$ of the vector $\vec{u}_A$. Express your answer using $\dot{\vec{u}}_C$, $\vec{\omega}$ and $\vec{r}$. 0.40

The speed of point A is given by the expression:

$$\vec{v}_A = \vec{v}_C + [\vec{\omega} \times \vec{r}].$$

Поскольку вектор $\vec{r}$ is perpendicular to the plane of the wall, we have:

Since the wall is flat, the vector $\vec{r}$ remains constant during the movement, therefore, after differentiation we have:

Answer:

$$\vec{u}_A = \vec{u}_C + [\vec{\omega} \times \vec{r}].$$

- A2** Determine the friction force $\vec{F}_0$ acting on the ball at the initial moment of contact with the wall. Express your answer in terms of $\vec{e}_x$, $\vec{e}_z$, α , μ and the normal reaction force of the wall N_0 at the initial moment. 0.60

At the initial moment we have:

$$\vec{u}_C(0) = v \sin \alpha \vec{e}_x \quad [\vec{\omega}(0) \times \vec{r}] = -v \cos \alpha \vec{e}_z \Rightarrow \vec{u}_A(0) = v (\sin \alpha \vec{e}_x - \cos \alpha \vec{e}_z).$$

Обратим внимание, что вектор $(\sin \alpha \vec{e}_x - \cos \alpha \vec{e}_z)$ является единичным. Сила трения $\vec{F}$ равна по модулю μN и направлена противоположно компоненте скорости $\vec{u}_A$, поэтому:

$$\vec{F} = -\frac{\vec{u}_A}{|\vec{u}_A|} \cdot \mu N,$$

or:

Answer:

$$\vec{F} = \mu N (\cos \alpha \vec{e}_z - \sin \alpha \vec{e}_x).$$

- A3** Prove that the time derivative $\dot{\vec{u}}_A$ of the velocity component $\vec{u}_A$ is related to the friction force $\vec{F}$ by the relation: 1.00

$$\dot{\vec{u}}_A = \frac{7\vec{F}}{2m}.$$

This fact can be used further, even if you could not prove it.

From the theorem on the motion of the center of mass for a ball it follows:

$$m\dot{\vec{u}}_C = \vec{F}.$$

Пусть $I = 2mR^2/5$ – момент инерции однородного шара относительно диаметра. Поскольку шар сферически симметричен, его момент импульса $\vec{L}_C$ относительно центра масс составляет:

$$\vec{L}_C = I\vec{\omega}.$$

Запишем основное уравнение динамики вращательного движения относительно центра масс шара:

$$\frac{d\vec{L}_C}{dt} = I\dot{\vec{\omega}} = \vec{M} = [\vec{r} \times \vec{F}].$$

$$\|\mathbf{b}\| \|\vec{r}\| \times \dot{\|\vec{\omega}\|} = \|\mathbf{b}\| \|\vec{r}\| \times \|\mathbf{F}\| \|\mathbf{b}\| = \|\vec{r}\| \|\mathbf{b}\| (\|\vec{r}\| \cdot \|\mathbf{F}\| \|\mathbf{b}\| - r^2 \|\mathbf{F}\| = -r^2 \|\mathbf{F}\| \Rightarrow \|\mathbf{F}\| = \frac{\|\mathbf{b}\| \dot{\|\vec{\omega}\|} \times \|\vec{r}\| \|\mathbf{b}\|}{r^2} \}.$$
$$\dot{\vec{u}}_A = \dot{\vec{u}}_C + \bigl[\dot{\vec{\omega}} \times \vec{r}\bigr] = \frac{\vec{F}}{m} + \vec{F}r^2$$

$$I = 7\vec{F}\cdot\vec{F} \quad \$ \$$$

The friction force vector $\vec{F}$ is directed against the velocity component $\vec{u}_A$ and codirected with the derivative of the velocity component $\dot{\vec{u}}_A$. It follows from this that the directions $\vec{u}_A$ and $\vec{F}$ are preserved during the collision. Then from point A3 we have:

$$\frac{du_A}{dt} = -\frac{7F}{2m} = -\frac{7\mu N}{2m} \Rightarrow u_{A_K} - u_A(0) = u_{A_K} - v = -\frac{7\mu}{2m} \int_0^t N dt.$$

$$\int_0^t N dt = \Delta p_y = 2mv \cos \alpha.$$
$$u_{A_K} = v(1 - 7\mu \cos \alpha),$$
$$\mu \leq \frac{1}{7 \cos \alpha}.$$

Answer:

$$\vec{u}_{Ak} = \begin{cases} v(1 - 7\mu \cos \alpha)(\sin \alpha \vec{e}_x - \cos \alpha \vec{e}_z) & \text{при } \mu \leq \frac{1}{7 \cos \alpha} \\ 0 & \text{при } \mu \geq \frac{1}{7 \cos \alpha} \end{cases}$$

$$\beta = \operatorname{arctg} \frac{v_{Cz}}{\sqrt{v_{Cx}^2 + v_{Cy}^2}}.$$
$$\vec{v}_C = v \cos \alpha \vec{e}_y + \vec{u}_C.$$
$$\vec{u}_C = \vec{u}_C(0) + \int_0^t \frac{\vec{F} dt}{m} = v \sin \alpha \vec{e}_x + \frac{2(\vec{u}_A - \vec{u}_A(0))}{7}$$

При $\mu \leq 1/(7 \cos \alpha)$ имеем:

$$\vec{u}_C = v \sin \alpha \vec{e}_x - \frac{2}{7} \cdot 7\mu v \cos \alpha (\sin \alpha \vec{e}_x - \cos \alpha \vec{e}_z) = v \sin \alpha (1 - 2\mu \cos \alpha) \vec{e}_x + 2\mu v \cos^2 \alpha \vec{e}_z.$$

Таким образом:

$$\vec{v}_C = v \sin \alpha (1 - 2\mu \cos \alpha) \vec{e}_x + v \cos \alpha \vec{e}_y + 2\mu v \cos^2 \alpha \vec{e}_z,$$

или же:

$$\beta = \arctg \frac{2\mu \cos^2 \alpha}{\sqrt{\cos^2 \alpha + \sin^2 \alpha (1 - 2\mu \cos \alpha)^2}}.$$

Если же $\mu \geq 1/(7 \cos \alpha) - \vec{u}_A = 0$. Тогда:

$$\vec{u}_C = v \sin \alpha \vec{e}_x - \frac{2v(\sin \alpha \vec{e}_x - \cos \alpha \vec{e}_z)}{7} = \frac{5v \sin \alpha \vec{e}_x}{7} + \frac{2v \cos \alpha \vec{e}_z}{7}.$$

Таким образом:

$$\vec{v}_C = \frac{5v \sin \alpha \vec{e}_x}{7} + v \cos \alpha \vec{e}_y + \frac{2v \cos \alpha \vec{e}_z}{7},$$

или же:

$$\beta = \arctg \frac{2 \cos \alpha}{\sqrt{(5 \sin \alpha)^2 + (7 \cos \alpha)^2}}.$$

Final:

Answer:

$$\beta = \begin{cases} \arctg \frac{2\mu \cos^2 \alpha}{\sqrt{\cos^2 \alpha + \sin^2 \alpha (1 - 2\mu \cos \alpha)^2}} & \text{при } \mu \leq \frac{1}{7 \cos \alpha} \\ \arctg \frac{2 \cos \alpha}{\sqrt{(5 \sin \alpha)^2 + (7 \cos \alpha)^2}} & \text{при } \mu \geq \frac{1}{7 \cos \alpha} \end{cases}$$

A6 At $\mu < \mu_{max}$, determine the coordinates x_C, y_C of the center of the ball at the moment it falls on the table. Express your answers using v, g, μ and α . 0.40

The ball will fall on the table surface after time t , equal to:

$$t = \frac{2v_{Cz}}{g}.$$

Тогда для координат x_C и y_C имеем:

$$x_C = v_{Cx}t = \frac{2v_{Cx}v_{Cz}}{g} \quad y_C = v_{Cy}t = \frac{2v_{Cy}v_{Cz}}{g}.$$

Подставляя v_{Cx}, v_{Cy} и v_{Cz} при разных значениях μ , we get:

Answer:

$$x_C = \begin{cases} \frac{4\mu v^2 \cos^2 \alpha \sin \alpha (1 - 2\mu \cos \alpha)}{g} & \text{при } \mu \leq \frac{1}{7 \cos \alpha} \\ \frac{10v^2 \sin 2\alpha}{49g} & \text{при } \mu \geq \frac{1}{7 \cos \alpha} \end{cases}$$

A7 For arbitrary values of μ , determine the amount of heat Q released during the collision of the ball with the wall. Express your answer using m, v, μ and α . Note: explicit calculation of the work done by the friction force will significantly simplify the solution of the problem. 1.00

Note: explicit calculation of the work done by the friction force will significantly simplify the solution of the problem.

For the power P_F of the friction force we have:

$$P_F = \vec{F} \cdot \vec{u}_A = \frac{2m\vec{u}_A \cdot \dot{\vec{u}}_A}{7}.$$

Тогда для количества теплоты Q имеем:

$$Q = - \int_0^t P_F dt = \int_{u_{Ak}}^{u_A(0)} \frac{2mu_A \dot{u}_A}{7} = \frac{m(u_A^2(0) - u_{Ak}^2)}{7}.$$

Учитывая, что $u_A(0) = v$, а также выражение для $u_{Ak}(\mu)$, we obtain:

Answer:

$$Q = \begin{cases} mv^2(2\mu \cos \alpha - 7\mu^2 \cos^2 \alpha) & \text{при } \mu \leq \frac{1}{7 \cos \alpha} \\ \frac{mv^2}{7} & \text{при } \mu \geq \frac{1}{7 \cos \alpha} \end{cases}$$

B1 Determine the velocity vector components of the ball's center v_φ and v_z in a cylindrical coordinate system. Express your answers using r , $\dot{\varphi}$ and $\dot{z}$. 0.20

The radius vector $\vec{r}_C$ of the center of the ball can be represented in the following form:

$$\vec{r}_C = r\vec{e}_r + z\vec{e}_z.$$

Дифференцируя:

$$\vec{v}_C = \frac{d\vec{r}_C}{dt} = r\dot{\vec{e}}_r + \dot{z}\vec{e}_z = r\dot{\varphi}\vec{e}_\varphi + \dot{z}\vec{e}_z.$$

Thus:

Answer:

$$v_\varphi = r\dot{\varphi} \quad v_z = \dot{z}.$$

B2 Determine the acceleration vector components of the ball's center a_r , a_φ and a_z in a cylindrical coordinate system. Express your answers using r , v_φ , $\dot{v}_\varphi$ and $\dot{v}_z$. 0.30

Let us differentiate the velocity vector with respect to time:

$$\vec{a}_C = \frac{d\vec{v}_C}{dt} = v_\varphi\dot{\vec{e}}_\varphi + \dot{v}_\varphi\vec{e}_\varphi + \dot{v}_z\vec{e}_z = -v_\varphi\dot{\varphi}\vec{e}_r + \dot{v}_\varphi\vec{e}_\varphi + \dot{v}_z\vec{e}_z.$$

Учитывая, что $v_\varphi = r\dot{\varphi}$, we obtain:

Answer:

$$a_r = -\frac{v_\varphi^2}{r} \quad a_\varphi = \dot{v}_\varphi \quad a_z = \dot{v}_z.$$

B3 From the condition of no slipping, determine the components of the angular velocity of the ball ω_φ and ω_z in the cylindrical coordinate system. Express your answers using r , v_φ and v_z . 0.40

For the speed of the point A of the ball at which it contacts the edge of the table, we have:

$$\vec{v}_A = \vec{v}_C + [\vec{\omega} \times \overrightarrow{CA}],$$

где $\vec{CA}$ – вектор, проведённый от центра C шара в точку A . Запишем условие равенства нулю скорости точки A шара, в которой он контактирует с краем стола:

$$v_{A\varphi} = v_\varphi - \omega_z r \quad v_{Az} = v_z + \omega_\varphi r.$$

Thus:

Answer:

$$\omega_z = \frac{v_\varphi}{r} \quad \omega_\varphi = -\frac{v_z}{r}.$$

C1 Determine the friction force $F_\varphi(\varphi)$ component acting on the ball, as well as the acceleration component $a_\varphi(\varphi)$ of its center. Express your answers in terms of the mass of the ball m , g and φ .

0.80

Let us write down the theorem on the motion of the center of mass for a ball in projection onto the axis directed along the vector e_φ :

$$ma_\varphi = F_\varphi + mg \sin \varphi.$$

Запишем уравнение динамики вращательного движения относительно оси z , проходящей через центр шара:

$$I_C \varepsilon_z = I_C \dot{\omega}_z = -r F_\varphi.$$

Поскольку $\dot{\omega}_z = \dot{v}_\varphi / r = a_\varphi / r$, после деления уравнений получим:

$$\frac{mr^2}{I_C} = -\frac{F_\varphi + mg \sin \varphi}{F_\varphi},$$

from:

Excluding F_φ , we get:

Answer:

$$F_\varphi = -\frac{mg \sin \varphi}{1 + mr^2 / I_C} = -\frac{2mg \sin \varphi}{7}.$$

C2 Get addicted $v_\varphi(\varphi)$. Express your answer in terms of v , g , r , α and φ .

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Multiplying the expression for v_φ and taking into account that $v_\varphi = r\dot{\varphi}$, we get:

$$a_\varphi v_\varphi = v_\varphi \dot{v}_\varphi = \frac{5gv_\varphi \sin \varphi}{7} = \frac{5gr \sin \varphi \dot{\varphi}}{7}.$$

Проинтегрируем полученное выражение по времени:

$$\int_{v \cos \alpha}^{v_\varphi(\varphi)} v_\varphi dv_\varphi = \int_0^\varphi \frac{5gr \sin \varphi d\varphi}{7} \Rightarrow \frac{v_\varphi^2 - v^2 \cos^2 \alpha}{2} = \frac{5gr(1 - \cos \varphi)}{7}.$$

Thus:

Answer:

$$v_\varphi = \sqrt{v^2 \cos^2 \alpha + \frac{10gr(1 - \cos \varphi)}{7}}.$$

C3 Under what condition does the ball not come off the table at the moment when the lowest point of the ball reaches its edge? Write this condition using v , g , r and α . In all subsequent paragraphs, assume that this condition is met.

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For the reaction force N at the moment when the lower point of the ball reaches the edge of the table, for the reaction force N we have:

$$N = mg - \frac{mv^2 \cos^2 \alpha}{r}.$$

Отрыва нет при условии $N \geq 0$, therefore:

Answer:

$$v \cos \alpha \leq \sqrt{gr}.$$

C4 Determine the angle φ_1 at the moment the ball leaves the table. Express your answer in terms of v , g , r and α .

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At the moment the ball comes off the table $N = 0$, therefore we have:

$$N = mg \cos \varphi_1 - \frac{mv_{\varphi_1}^2}{r}.$$

Приравняем два выражения для v_{φ}^2 :

$$v_{\varphi_1}^2 = gr \cos \varphi_1 = v^2 \cos^2 \alpha + \frac{10gr(1 - \cos \varphi_1)}{7}.$$

Таким образом:

$$\cos \varphi_1 = \frac{10gr + 7v^2 \cos^2 \alpha}{17gr},$$

откуда для φ_1 we find:

Answer:

$$\varphi_1 = \arccos \left(\frac{10}{17} + \frac{7v^2 \cos^2 \alpha}{17gr} \right).$$

D1 Express the kinetic energy of the ball E_k in terms of m , v_{φ} , v_z , ω_r and r .

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Let's use Koenig's theorem:

$$E_k = \frac{mv_C^2}{2} + \frac{I_C \omega^2}{2} = \frac{mv_{\varphi}^2}{2} + \frac{mv_z^2}{2} + \frac{I_C \omega_r^2}{2} + \frac{I_C \omega_{\varphi}^2}{2} + \frac{I_C \omega_z^2}{2}.$$

Подставляя ω_{φ} и ω_z , we get:

Answer:

$$E_k = \left(m + \frac{I_C}{r^2} \right) \frac{v_{\varphi}^2}{2} + \left(m + \frac{I_C}{r^2} \right) \frac{v_z^2}{2} + \frac{I_C \omega_r^2}{2} = \frac{7m(v_{\varphi}^2 + v_z^2)}{10} + \frac{mr^2 \omega_r^2}{5}.$$

D2 Write down the law of conservation of mechanical energy for the ball. Combining it with the result of item C2, show that the quantities ω_r and v_z are related by the relation:

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$$1 = \frac{\omega_r^2}{A^2} + \frac{v_z^2}{B^2},$$

где $A, B > 0$ — постоянные коэффициенты. Определите A и B . Ответы выразите через v , r и α .

The law of conservation of energy is as follows:

$$E_k = \frac{7m(v_{\varphi}^2 + v_z^2)}{10} + \frac{mr^2 \omega_r^2}{5} = E_{k0} + A_{\text{тяж}} = \frac{7mv^2}{10} + mgr(1 - \cos \varphi).$$

откуда:

$$v_\varphi^2 = v^2 + \frac{10gr(1 - \cos \varphi)}{7} - v_z^2 - \frac{2\omega_r^2 r^2}{7}.$$

Подставляя зависимость $v_\varphi^2(\varphi)$, получим:

$$v^2 \cos^2 \alpha + \frac{10gr(1 - \cos \varphi)}{7} = v^2 + \frac{10gr(1 - \cos \varphi)}{7} - v_z^2 - \frac{2\omega_r^2 r^2}{7},$$

откуда:

$$v_z^2 + \frac{2\omega_r^2 r^2}{7} = v^2 \sin^2 \alpha \Rightarrow 1 = \frac{v_z^2}{v^2 \sin^2 \alpha} + \frac{2\omega_r^2 r^2}{7v^2 \sin^2 \alpha}.$$

Для A и B we have:

Answer:

$$A = \sqrt{\frac{7}{2}} \frac{v \sin \alpha}{r} \quad B = v \sin \alpha.$$

D3 The angular acceleration vector $\vec{\varepsilon}$ of the ball can be represented as:

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$$\vec{\varepsilon} = \varepsilon_r \vec{e}_r + \varepsilon_\varphi \vec{e}_\varphi + \varepsilon_z \vec{e}_z.$$

Используя уравнение динамики вращательного движения относительно центра шара, покажите, что $\varepsilon_r = 0$.
Используя полученное равенство, выразите $\dot{\omega}_r$ через $\dot{\varphi}$, v_z и r .

From the equation of dynamics of rotational motion relative to the center of the ball we have:

$$\frac{d\vec{L}_C}{dt} = I\vec{\varepsilon} = [\vec{CA} \times (\vec{F} + \vec{N})].$$

Поскольку $\vec{N} \parallel \vec{CA}$, имеем:

$$I\vec{\varepsilon} = [\vec{CA} \times \vec{F}].$$

Let's use the expression for angular acceleration in a cylindrical coordinate system:

$$\varepsilon_r = \dot{\omega}_r - \dot{\varphi}\omega_\varphi = 0,$$

from:

Answer: Since $\vec{\varepsilon} \perp \vec{CA}$, the angular acceleration component is $\varepsilon_r = 0$.

D4 Combining the results of points D2 and D3, you obtain dependencies $\omega_r(\varphi)$ and $v_z(\varphi)$. Express your answers using v , α , r and φ . **1.20**

Let's use the result of point B4:

$$\dot{\omega}_r = -\frac{\dot{\varphi}v_z}{r}.$$

Выражая v_z и подставляя в уравнение, полученное в предыдущем пункте, находим:

$$1 = \frac{\omega_r^2}{A^2} + \frac{r^2 \dot{\omega}_r^2}{B^2 \dot{\varphi}^2}.$$

Обратим внимание, что это уравнение с разделяющимися переменными ω_r и φ :

$$d\varphi = -\frac{r}{B} \frac{d\omega_r}{\sqrt{1 - \frac{\omega_r^2}{A^2}}}. \quad \text{Вводя переменную } t = \omega_r/A \text{ и интегрируя, находим :}$$

$$\dot{\varphi} = -\frac{rA}{B} \int \lim_{t \rightarrow 0} \frac{dt}{\sqrt{1-t^2}} = -\frac{rA}{B} \arcsin t \Big|_0^{\varphi} \Rightarrow \dot{\varphi} = -\sin \left(\frac{B\varphi}{rA} \right),$$

или же :

$$\omega_r(\varphi) = -A \sin \left(\frac{B\varphi}{rA} \right).$$

Подставляя $t(\varphi)$ в уравнение, связывающее v_z с ω_r , находим :

$$\dot{v}_z^2 = 1 - t^2 = \cos^2 \left(\frac{B\varphi}{rA} \right),$$

или же :

$$v_z = B \cos \left(\frac{B\varphi}{rA} \right), \quad \text{поскольку } v_z(0) > 0. \text{ Finally:}$$

Answer:

$$\omega_r(\varphi) = -\sqrt{\frac{7}{2}} \frac{v \sin \alpha}{r} \sin \left(\sqrt{\frac{2}{7}} \varphi \right) \quad v_z = v \sin \alpha \cos \left(\sqrt{\frac{2}{7}} \varphi \right).$$

D5 Let us consider the passage to the limit when the angle is $\alpha \rightarrow \pi/2$, i.e. the movement of the ball until it contacts the edge of the table occurs almost parallel to it. Determine the projection of the velocity v_z of the center of the ball, as well as the projection of its angular velocity ω_y onto the axis y , directed vertically downward, at the moment the ball lifts off the table. Express your answers using v and r . All numerical coefficients in the answer must be analytical, not approximate! 0.80

At $\alpha = \pi/2$ for φ_1 we have:

$$\varphi_1 = \arccos \left(\frac{10}{17} \right).$$

The projection of the angular velocity of the ball onto the axis y , directed vertically downward, is equal to:

$$\omega_y = \omega_\varphi \sin \varphi_1 - \omega_r \cos \varphi_1.$$

Подставляя ω_r , ω_φ , получим:

$$\omega_y = \frac{v}{r} \left(\sqrt{\frac{7}{2}} \cos \varphi_1 \sin \left(\sqrt{\frac{2}{7}} \varphi_1 \right) - \sin \varphi_1 \cos \left(\sqrt{\frac{2}{7}} \varphi_1 \right) \right).$$

После подстановки φ_1 we find:

Answer: Then for speed v_z we find:

$$v_z = v \cos \left(\sqrt{\frac{2}{7}} \arccos \left(\frac{10}{17} \right) \right).$$